CIS437 - Cloud Computing

– Serverless Functions –

Event-Driven Computing with Cloud Run Functions

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Based on the original version by Professor Erik Fredericks

Last Class

- Monolithic applications and their limitations
- Microservices architecture
- Benefits and challenges
- Decoupling and service boundaries
- When microservices make sense
- Monolith-to-Microservices lab

Last Class Practice



For each scenario, decide whether you would keep a monolith, move toward microservices, or use a mixed approach. Briefly explain why.

1. A small student project has three developers, a few hundred users, and a simple web application with login, posts, and comments.

2. An e-commerce application has Product, Order, Payment, and Shipping functions. Product traffic grows rapidly during sales, while the other functions remain relatively stable.

3. A large application has several teams. Payment changes frequently and has different security and scaling requirements, while most other functionality is stable.

What factor is driving your architecture choice?

Agenda

- Serverless Computing and FaaS
- How Serverless Functions Work
- Triggers and Common Use Cases
- Benefits, Trade-offs, and Cold Starts
- Function Design and Access Control
- Event-Driven Functions
- Serverless Cost
- Practice and Summary

What Is Serverless / FaaS?

Serverless computing

- Cloud provider manages the underlying infrastructure
- Resources are provisioned and scaled automatically

Function as a Service (FaaS)

- A type of serverless computing
- Runs small, focused functions in response to requests or events

Examples

- Google Cloud Run functions
- AWS Lambda
- Microsoft Azure Functions

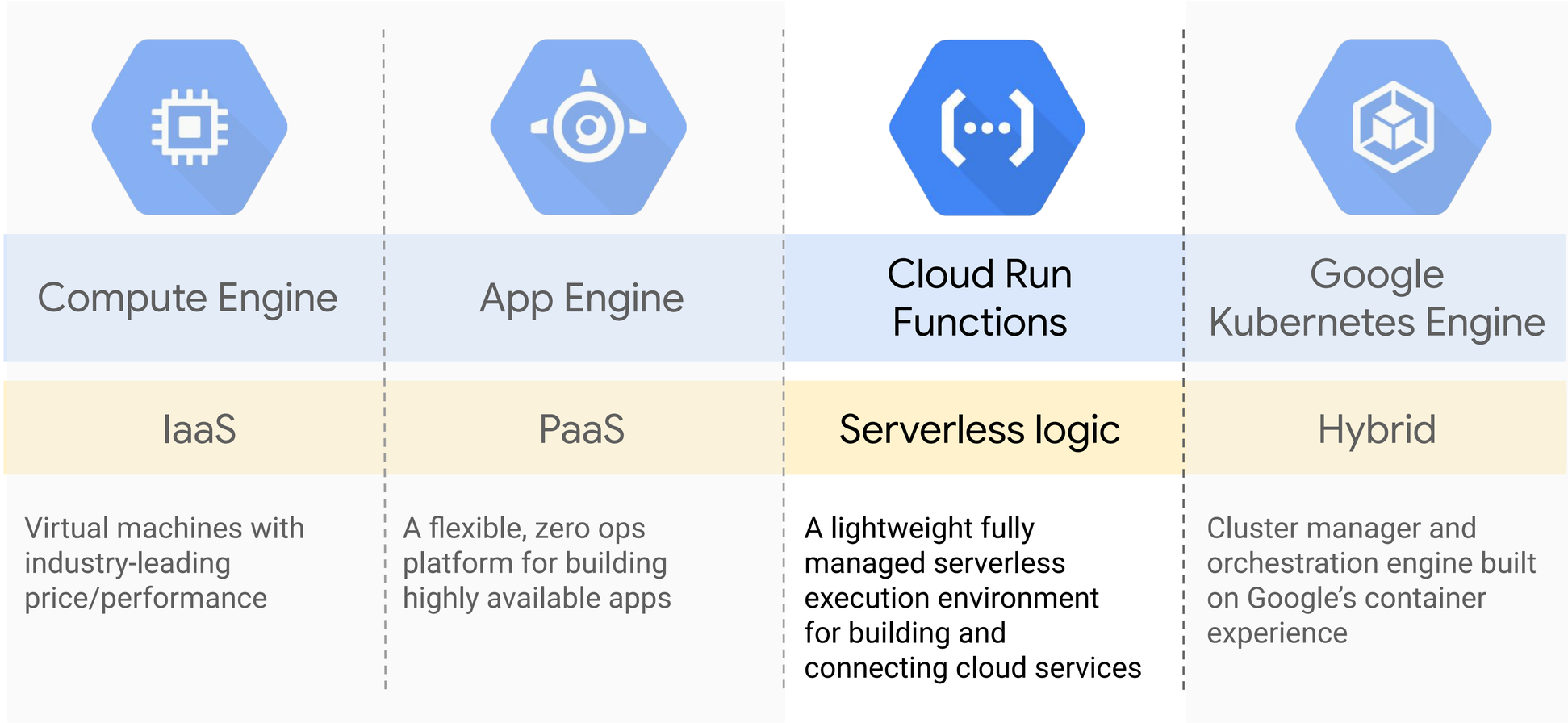
Can a Microservice Be a Serverless Function?



Can a microservice be a serverless function?

Yes — if the function represents a focused, independently deployable service responsibility.

Where Functions Fit in Cloud Computing



More infrastructure control ← Compute Engine — GKE — App Engine — Cloud Run functions → **Less infrastructure management**

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What Do You Manage with FaaS?

With FaaS:

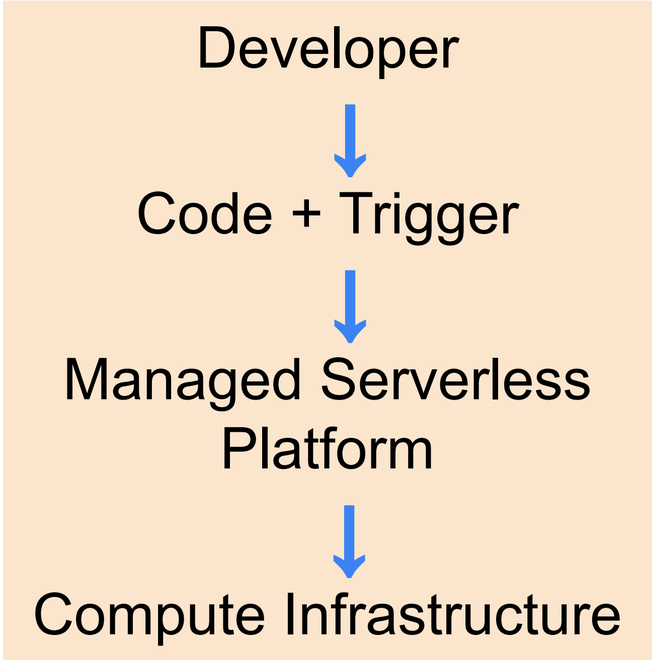
- You do not provision or manage servers
- The cloud provider manages the OS and runtime infrastructure
- You provide the function code and dependencies
- You configure triggers, access, resources, and scaling

Serverless Does Not Mean No Servers

Servers still exist.

You manage: code, dependencies, triggers, access

Cloud provider manages: servers, OS, runtime infrastructure, scaling



What Makes a Serverless Function Work?

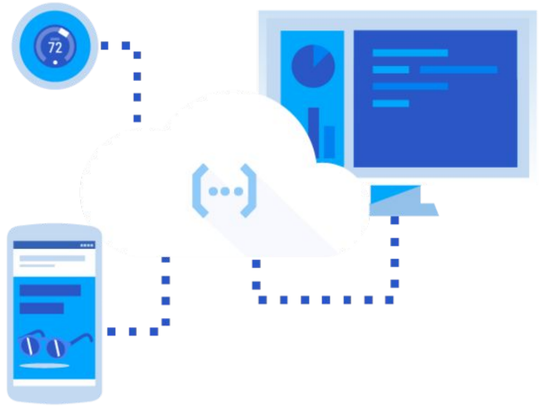
Code — what the function does **Cloud Services / APIs** — what the function can interact with
Trigger / Event — when the function runs **Managed Runtime** — where the function runs



Connect and extend cloud services



Events and triggers

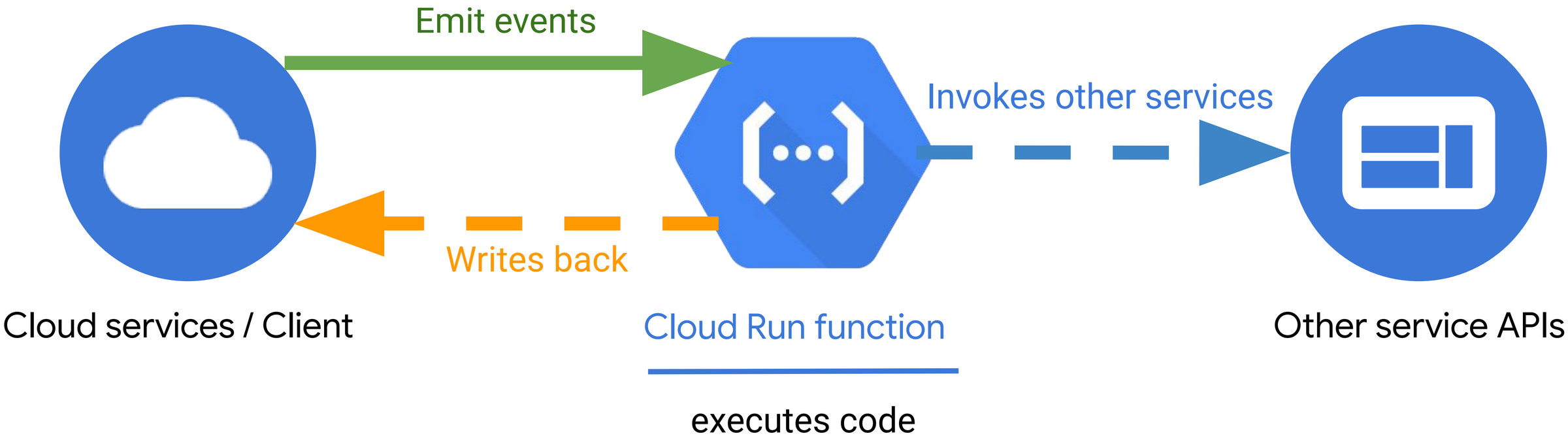


Serverless

Event → Trigger → Function → Action

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How Serverless Functions Work



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Event Triggers

Different triggers other than HTTPS!

Create triggers from Cloud Storage events

- Pub/Sub, some event
- Here, change in a cloud storage bucket

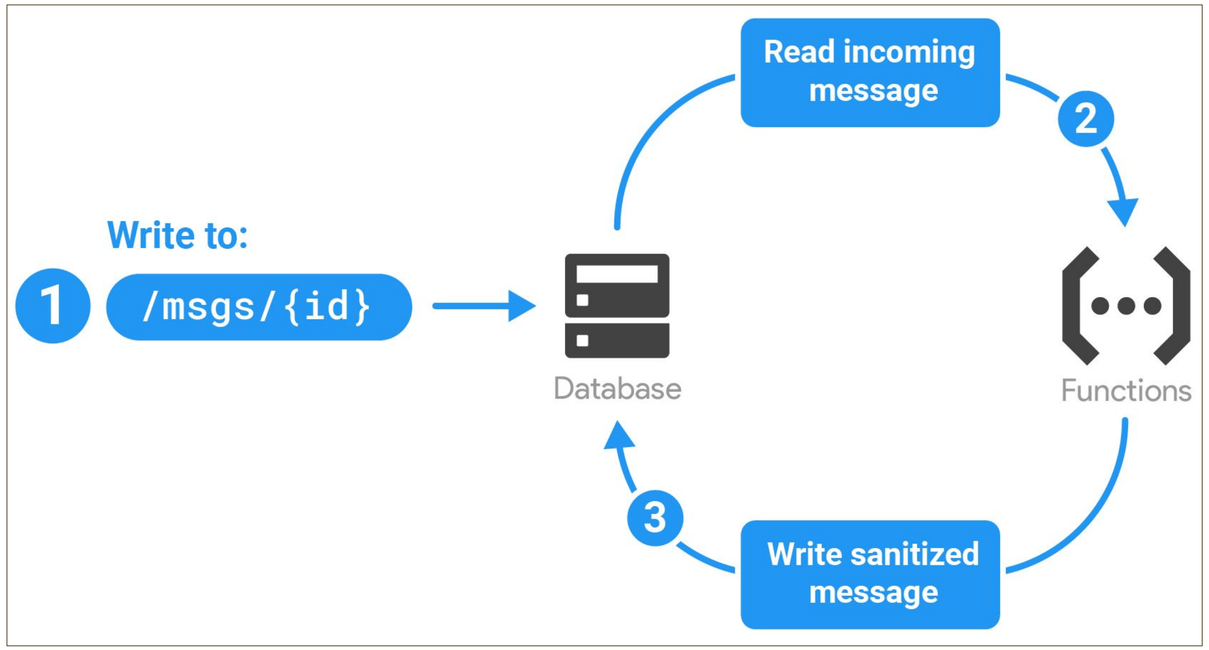


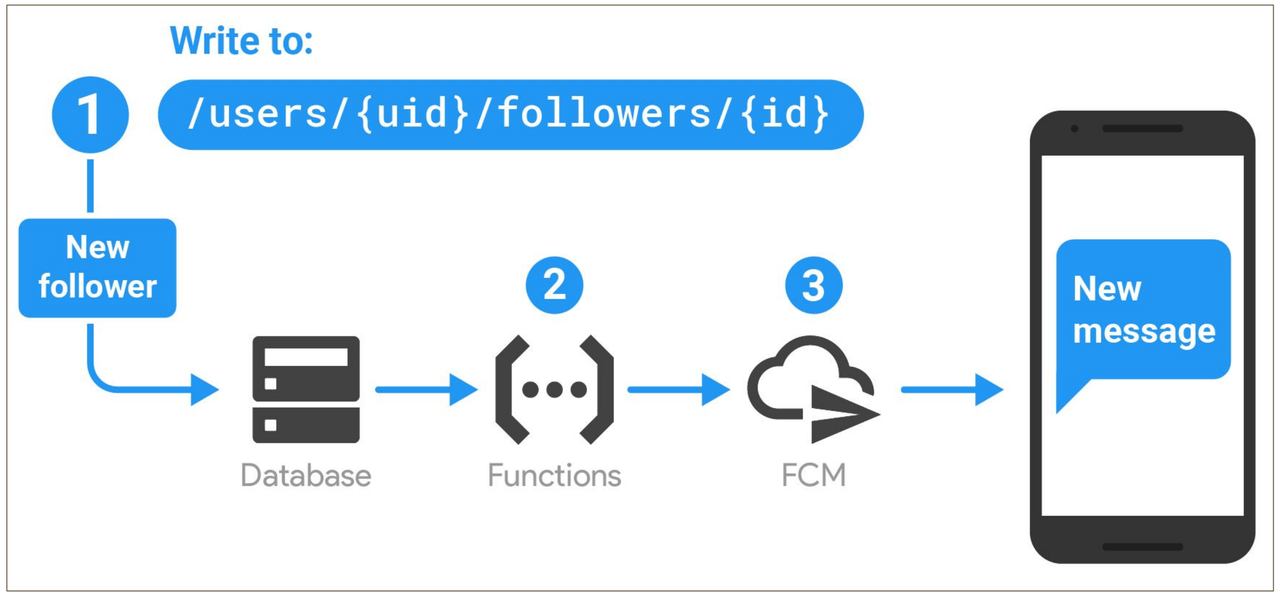
- For Cloud Storage events, keep the bucket, Eventarc trigger, and function in compatible locations; using the same region is the simplest setup for this class.

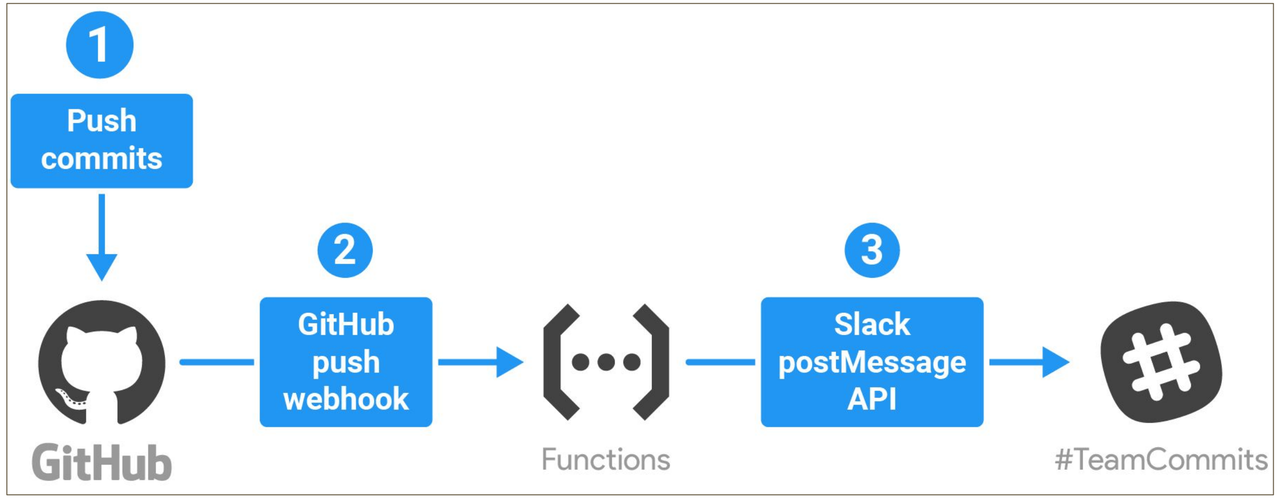
File upload → Cloud Storage event → Trigger → Cloud Run function → Logs / processing

Common Serverless Use Cases

- Data-change processing
- Notifications
- Webhooks and automation
- Background/event-driven tasks







What can you do with Cloud Functions?

Why Use Serverless Functions?

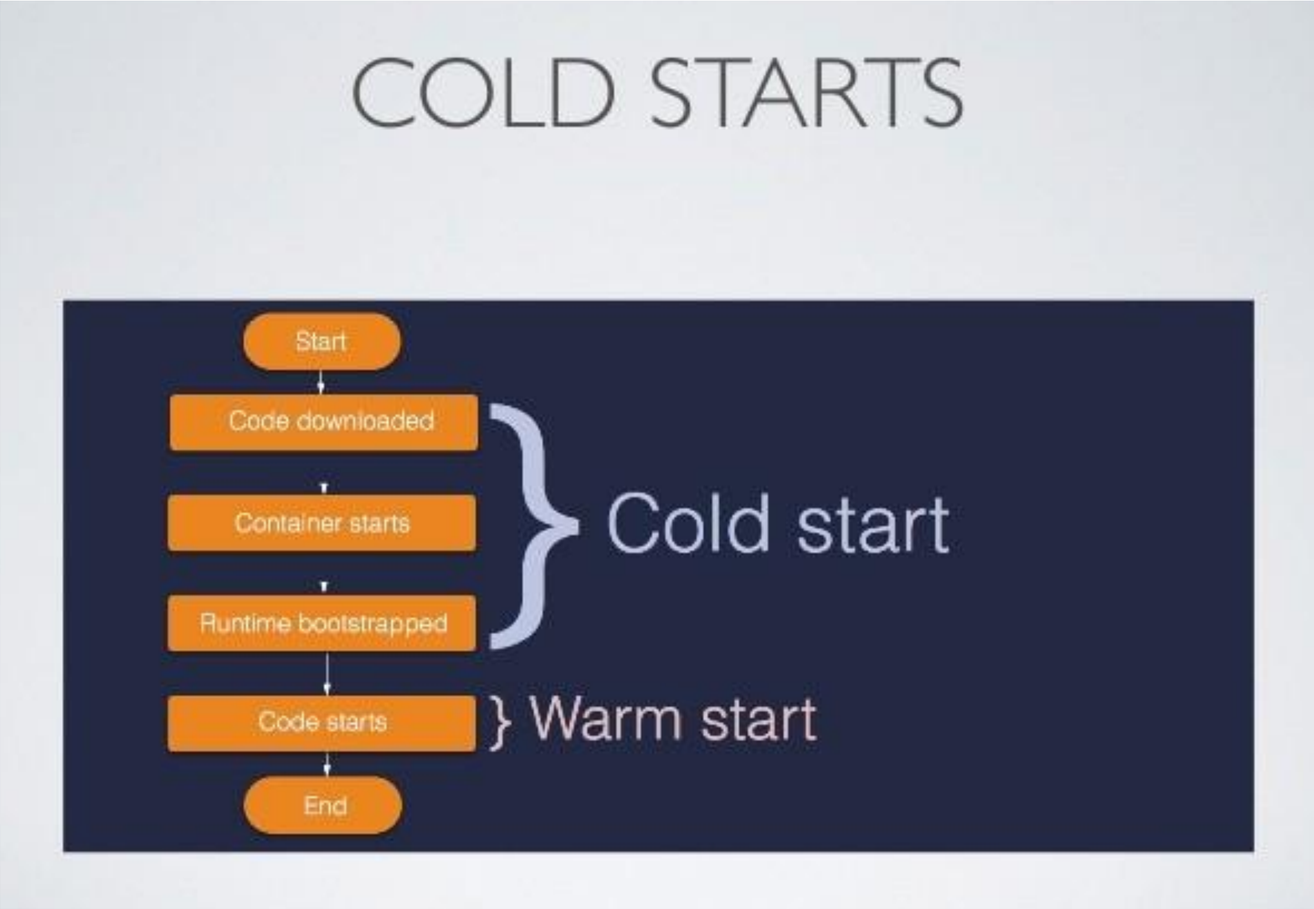
- No server management
- Automatic scaling
- Event-driven execution
- Fast deployment for small, focused tasks
- Can scale down when idle
- Usage-based pricing

Trade-offs and Limits

- Execution/resource limits
- Cold-start latency
- More difficult debugging and monitoring
- Dependency and package-size overhead
- Cost can increase at high volume
- Platform-specific configuration

Cold Start vs. Warm Start

- A function needs an execution environment before it can run
- **Cold start:** the platform must initialize a new execution environment
- **Warm start:** an existing environment is reused
- Cold starts can add latency to the request
- Minimum instances can reduce cold starts, but may increase cost



Function Design Best Practices

Design for retries / idempotency

- Repeating the same operation should not create unintended duplicate effects
- Important because event-driven functions may be invoked more than once

HTTP functions must return a response

- Otherwise the request may continue until timeout

Use dependencies carefully

- Avoid unnecessary libraries
- Dependencies can increase deployment size and cold-start latency

Do not assume local state persists

- A later invocation may run in a different execution environment

Functions best practices

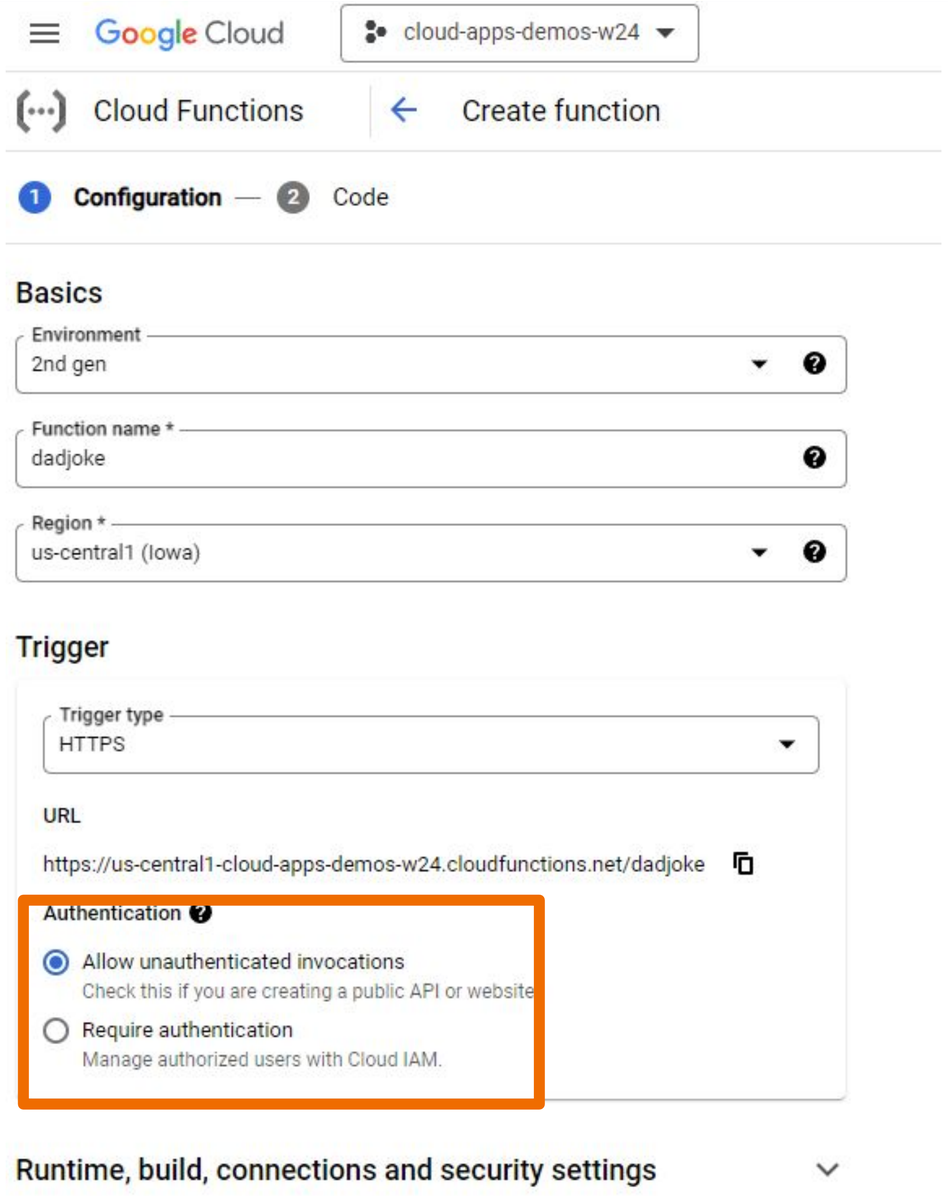
HTTP Functions: Public or Authenticated?

Public / unauthenticated

- Anyone who knows the endpoint can invoke it
- Useful for public APIs, webhooks, or public-facing services
- Requires careful input validation and abuse protection

Authenticated

- Only authorized users or services can invoke it
- Appropriate for internal APIs or sensitive operations
- Requires identity and permission management



Develop and Test Functions Locally

Functions Framework allows you to:

- run functions locally
- test before deployment
- debug more easily
- use the same function programming model locally and in the cloud

Write → Test locally → Debug → Deploy

functions-framework-python

Local functions development

Locally debug functions with AWS SAM

Event-Driven Example: File Upload

File uploaded to Cloud Storage



Event



Cloud Run function



Process / analyze file 



Save result / call another API

- Clean or validate the file
- Parse or transform the file
- Detect and flag bad/illegal uploads
- Send a notification
- Run AI/ML analysis

Where Could We Use This?

When a file is uploaded to Cloud Storage, a function could:

- Generate image thumbnails
- Scan or validate uploaded files
- Analyze images with AI/ML
- Extract or transform data
- Run ETL processing
- Send notifications when processing is complete

Common setup issue:

The Cloud Storage service agent may need permission to publish events through Pub/Sub/Eventarc.

How Does Serverless Cost Work?

Serverless cost is affected by:

- Number of requests/invocations
- CPU and memory used
- Execution duration
- Networking/data transfer
- Minimum instances, if configured

Key idea: Serverless can be inexpensive for intermittent workloads, but very high-volume workloads can still become expensive.

Cloud Run pricing

AWS Lambda pricing

Summary

- Serverless means the provider manages the underlying infrastructure
- FaaS runs focused code in response to requests or events
- Functions can be triggered by HTTP, Pub/Sub, Cloud Storage, and other events
- Serverless reduces infrastructure management and can scale automatically
- Trade-offs include cold starts, limits, debugging complexity, and cost
- Design functions to handle retries and duplicate events safely
- Cloud Run functions is Google Cloud's current function platform

Practice

Complete the practice questions on Blackboard

